

would then push back your time with the CPU and hand over the next meeting slot to some other process. This is, of course, an unfair allocation of CPU cycles.

Now, each modern process has a cycle counter register (a small piece of silicon), which keeps track of what a thread is doing with CPU cycles. Vista makes use of this register to track precisely how many CPU cycles a thread executes. It also makes an estimate of how many cycles a CPU can execute in a clock interval, which helps the kernel to accurately ensure that each application is getting the turn it requires at the CPU. The unfair manner in which the kernel might push back a process' scheduled go at the CPU is also alleviated: Vista does not count its own interrupt execution against a processes turn. This ensures that under Vista, a thread will always get a turn at the CPU.

3.2 Multimedia Class Scheduler Service

Vista also handles multimedia tasks in a neat manner. Today, there are many potentially CPU-intensive tasks always running in the background in an operating system. Threads from programs such as anti-virus scanners, e-mail clients, instant messengers, search services, content indexers and so on constantly vie for CPU cycles. So much so that you might experience audio skipping or video stuttering as these tasks take over your CPU cycles. Vista offers a means to ensure that media playback remains a smooth experience even as it balances and juggles these myriad non-enter-taining but necessary tasks.

This act of juggling is primarily done through the MMCSS service introduced briefly above. Under Vista, a multimedia application can register itself with MMCSS, telling the service about its multimedia needs on a task-wise basis. This registering happens in the Windows registry under the `HKEY_LOCAL_MACHINE\Software\Microsoft\Windows NT\Currentversion\Multimedia\`